

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4617: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Consider that a game of Pacman is being played on the grid shown in Figure 1.

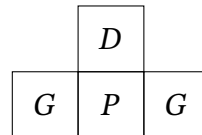


Figure 1: Pacman Grid for Question 1

Here, P indicates the position of Pacman, G indicates the position of the ghosts, and D indicates the position of a food dot. For simplicity, the ghosts remain stationary throughout the game.

To determine the policy for playing the game, our feature-based Q-learning agent, Pacman, uses two features, F_g and F_d defined as follows:

$$F_g(s, a) = F_1(s) + F_2(s, a) + F_3(s, a)$$

$$F_d(s, a) = F_4(s) + 2 \times F_5(s, a)$$

where

$F_1(s)$ = number of ghosts within 1 step of state s

$F_2(s, a)$ = number of ghosts Pacman touches after taking action a from state s

$F_3(s, a)$ = number of ghosts within 1 step of the state Pacman ends up in after taking action a

$F_4(s)$ = number of food dots within 1 step of state s

$F_5(s, a)$ = number of food dots eaten after taking action a from state s

After a few episodes of Q-learning, the weights are $w_g = -10$ and $w_d = 100$. The discount factor, $\gamma = 0.5$ and the learning rate, $\alpha = 0.5$. The action space of Pacman is $\{left, right, up, down, stay\}$. Pacman can take any actions from a state given it does not go beyond the grid.

- a) Considering the actions, a that are available from the current position, s of Pacman, answer the following: 16 + 8
(CO1)
(PO1)
- i. Calculate $F_g(s, a)$ and $F_d(s, a)$.

Solution:

$$F_g(s, left) = 2 + 1 + 1 = 4$$

$$F_g(s, right) = 2 + 1 + 1 = 4$$

$$F_g(s, up) = 2 + 0 + 0 = 2$$

$$F_g(s, stay) = 2 + 0 + 2 = 4$$

$$F_d(s, left) = 1 + 2 \times 0 = 1$$

$$F_d(s, right) = 1 + 2 \times 0 = 1$$

$$F_d(s, up) = 1 + 2 \times 1 = 3$$

$$F_d(s, stay) = 1 + 2 \times 0 = 1$$

$$F_g(s, left) = 4 \quad F_d(s, left) = 1$$

Rubric:

- 1 point for each correct equation.
- 1 point for each correct value.

ii. Calculate the approximate Q-value of the state-action pair (s,a) , $\hat{Q}(s,a)$.

Solution:

$$\begin{aligned}\hat{Q}(s, up) &= w_g F_g(s, up) + w_d F_d(s, up) = (-10) \times 2 + 100 \times 3 = 280 \\ \hat{Q}(s, left) &= w_g F_g(s, left) + w_d F_d(s, left) = (-10) \times 4 + 100 \times 1 = 60 \\ \hat{Q}(s, right) &= w_g F_g(s, right) + w_d F_d(s, right) = (-10) \times 4 + 100 \times 1 = 60 \\ \hat{Q}(s, stay) &= w_g F_g(s, stay) + w_d F_d(s, stay) = (-10) \times 4 + 100 \times 1 = 60\end{aligned}$$

Rubric:

- 1 points for each correct equation.
- 1 point for each Q-Value.

b) Recommend the optimal policy for Pacman from its current position following the $\hat{Q}(s,a)$ values that you calculated in 1.a) ii. Argue on how good the policy is considering the alternatives.

3 + 3
(CO3)
(PO3)

Solution:

The optimal policy for the current position of Pacman should be:

$$\begin{aligned}\pi(s) &= \operatorname{argmax} (\hat{Q}(s, up), \hat{Q}(s, left), \hat{Q}(s, right), \hat{Q}(s, stay)) \\ &= up\end{aligned}$$

Since going up results in Pacman eating the food dot, it is reasonable to decide to go up. Compared to other moves, where Pacman stays in the same position or gets eaten by the ghost, this is the best possible outcome for Pacman.

Rubric:

- 1 point for the equation
- 2 points for the correct decision.
- 3 points for the argument.

c) From its current position, s , Pacman moves up to go to the cell, s' containing the food dot and eats it. We observe a reward, $R(s, a, s') = 250$. Considering the actions, a' that are available from s' , answer the followings:

6 + 4
(CO1)
(PO1)

i. Calculate the exact Q-value of the state-action pair (s, up) , $Q(s, up)$. Use the approximate Q-values calculated in Question 1.a)ii if necessary.

Solution:

$$\begin{aligned}Q(s, up) &= R(s, a, s') + \gamma \times \max_{a'} \hat{Q}(s', a') \\ &= 250 + 0.5 \times \max (\hat{Q}(s', down), \hat{Q}(s', stay)) \\ &= 250 + 0.5 \times 0 \\ &= 250\end{aligned}$$

where

$$\hat{Q}(s', down) = w_g F_g(s', down) + w_d F_d(s', down) = (-10) \times 2 + 100 \times 0 = -20$$

$$\hat{Q}(s', stay) = w_g F_g(s', stay) + w_d F_d(s', stay) = (-10) \times 0 + 100 \times 0 = 0$$

Rubric:

- 1 point for the correct equation.
- 1 point for each of the correct $Q(s', a')$ equation.
- 1 point for each correct $Q(s', a')$ value.
- 1 point for the final result.

ii. Update w_g and w_d .

Solution:

$$w_g = w_g + \alpha \times (Q(s, up) - \hat{Q}(s, up)) \times F_g(s, a) = -10 + 0.5 \times (250 - 280) \times 2 = -40$$

$$w_d = w_d + \alpha \times (Q(s, up) - \hat{Q}(s, up)) \times F_d(s, a) = 100 + 0.5 \times (250 - 280) \times 3 = 55$$

Rubric:

- 1 points for each equation.
- 1 point for each correct value.

2. a) Consider a Markov Decision Process (MDP) having three states S_1 , S_2 , and S_3 , with rewards -1 , -2 , and 0 for executing an action from the state, respectively. S_3 is the terminal state where only the exit action is available that gives no reward. There is no other living reward. S_1 and S_2 each have two possible actions: a and b . 12 × 2
(CO3)
(PO3)

The transition function is described as follows:

- In S_1 , action a moves the agent to S_2 with probability 0.8 and makes the agent stay in S_1 , otherwise.
- In S_2 , action a moves the agent to S_1 with probability 0.8 and makes the agent stay in S_2 , otherwise.
- In either S_1 or S_2 , action b moves the agent to S_3 with probability 0.1 and makes the agent stay in its current state, otherwise.

Assume that the discount factor, $\gamma = 1$. With detailed steps, recommend the optimal policy for each state considering:

- i. The initial policy has action b in both states.

Solution:

Initially, $\pi(S) = b$ for $S \in \{S_1, S_2\}$. $R(S_1) = -1$, $R(S_2) = -2$, and $R(S_3) = 0$.

Policy Evaluation:

$$V(S_1) = -1 + 0.1 \times V(S_3) + 0.9 \times V(S_1)$$

$$V(S_2) = -2 + 0.1 \times V(S_3) + 0.9 \times V(S_2)$$

$$V(S_3) = 0$$

We get: $V(S_1) = -10$ and $V(S_2) = -20$.

Policy Improvement:

$$Q(S_1, a) = 0.8 \times V(S_2) + 0.2 \times V(S_1) \\ = -18$$

$$Q(S_1, b) = 0.1 \times V(S_3) + 0.9 \times V(S_1) \\ = -9$$

$$Q(S_2, a) = 0.8 \times V(S_1) + 0.2 \times V(S_2) \\ = -12$$

$$Q(S_2, b) = 0.1 \times V(S_3) + 0.9 \times V(S_2) \\ = -18$$

So the optimal policy:

$$\pi(S_1) = \operatorname{argmax} (Q(S_1, a), Q(S_1, b)) \\ = b$$

$$\pi(S_2) = \operatorname{argmax} (Q(S_2, a), Q(S_2, b)) \\ = a$$

Since the policy changed, we do another iteration:

Policy Evaluation:

$$V(S_1) = -1 + 0.1 \times V(S_3) + 0.9 \times V(S_1)$$

$$V(S_2) = -2 + 0.8 \times V(S_1) + 0.2 \times V(S_2)$$

$$V(S_3) = 0$$

We get: $V(S_1) = -10$ and $V(S_2) = -12.5$.

Policy Improvement:

$$Q(S_1, a) = 0.8 \times V(S_2) + 0.2 \times V(S_1) \\ = -12$$

$$Q(S_1, b) = 0.1 \times V(S_3) + 0.9 \times V(S_1) \\ = -9$$

$$Q(S_2, a) = 0.8 \times V(S_1) + 0.2 \times V(S_2) \\ = -19.25$$

$$Q(S_2, b) = 0.1 \times V(S_3) + 0.9 \times V(S_2) \\ = -11.25$$

So the optimal policy:

$$\pi(S_1) = \operatorname{argmax} (Q(S_1, a), Q(S_1, b)) \\ = b$$

$$\pi(S_2) = \operatorname{argmax} (Q(S_2, a), Q(S_2, b)) \\ = a$$

Since the policy remain unchanged, we can terminate. We get: $\pi(S_1) = b, \pi(S_2) = a, \pi(S_3) = EXIT$.

Rubric:

- 0.5 points for each value
- 0.5 points for each Q-value
- 0.5 points for each policy
- 1 point for each final policy

- ii. The initial policy has action a in both states.

Solution:

Initially, $\pi(S) = a$ for $S \in \{S_1, S_2\}$. $R(S_1) = -1$, $R(S_2) = -2$, and $R(S_3) = 0$.

Policy Evaluation:

$$V(S_1) = -1 + 0.2 \times V(S_1) + 0.8 \times V(S_2)$$

$$V(S_2) = -2 + 0.8 \times V(S_1) + 0.2 \times V(S_2)$$

$$V(S_3) = 0$$

We notice that $V(S_1)$ and $V(S_2)$ are inconsistent. If we were to try to solve them iteratively, we would find the values tending to $-\infty$. So this problem is unsolvable.

Rubric:

- 3 points for correct equations of V .
- 7 points for identifying that it goes to negative infinity.
- 2 points for decision.

- b) Sometimes MDPs are formulated with a reward function $R(s, a)$ that depends on the current state and the action taken, or $R(s, a, s')$ that also depends on the outcome state. Modify the Bellman Equation to determine the value of a state based on these formulations.

6
(CO2)
(PO2)

Solution:

For $R(s, a)$, we have:

$$V(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \right]$$

And for $R(s, a, s')$, we have:

$$V(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s')]$$

Rubric:

- 3 points for each formulation.

- c) Suppose that we define the value of a state to be the maximum (as opposed to summation) reward obtained from its future states. Does this result in stationary preferences? Justify your position.

2 + 6
(CO1)
(PO1)

Solution:

Stationary preferences require the agent to follow:

$$[a_1, a_2, \dots] \succ [b_1, b_2, \dots]$$

$$\Updownarrow$$

$$[r, a_1, a_2, \dots] \succ [r, b_1, b_2, \dots]$$

However, if the utility of a sequence is its maximum reward, we can easily violate the stationary preferences. For example,

$$[4, 3, 0, 0, \dots] \sim [4, 0, 0, 0, \dots]$$

but if we remove the first item from both sequences:

$$[3, 0, 0, \dots] \succ [0, 0, 0, \dots]$$

So it breaks stationary preferences.

Rubric:

- 2 points for decision.
- 3 points for the theorem.
- 3 points for the counterexample.

3. Consider the problem of solving two 8-puzzles. A sample game is shown in Figure 2:

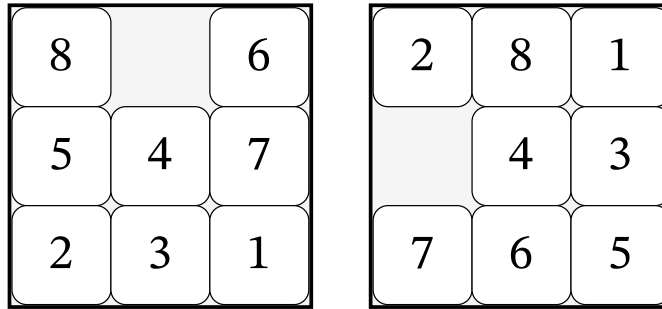


Figure 2: Sample 8-puzzle game for Question 3

- a) Formulate the scenario as a game by identifying the start state, action, successor function, and goal test.

8
(CO2)
(PO2)

Solution:

Start State: Two arbitrary 8-puzzle states

Action: Move one empty piece of one puzzle horizontally or vertically

Successor function: one move on an unsolved puzzle

Goal test: Both puzzles are solved

Rubric:

- 2 point for each

- b) Determine the size of the reachable state space.

3
(CO1)
(PO1)

Solution:

Each puzzle has $9!/2$ reachable states. The joint state space has $(9!)^2/4$ reachable states

Rubric:

- 1 point for single puzzle
- 2 point for both

- c) Suppose we make the problem adversarial by considering two players who take turns moving. Before each move, a coin is flipped to determine the puzzle on which to make a move in that turn. The winner is the first to solve one puzzle.

4 + 6
(CO3)
(PO3)

- i. With proper justification, recommend an algorithm for finding the optimal move in this game.

Solution:

Expectiminimax. Because there is chance involved.

Rubric:

- 2 points for choosing the correct one

- 2 points for justification

- ii. “The game will never end if both players play perfectly.” - Do you agree with the statement? Justify your position.

Solution:

I agree.

Consider a state in which the coin has come up heads, say, and you get to work on a puzzle that is 2 steps from the goal. Should you move one step closer? If you do, your opponent wins if he tosses heads; or if he tosses tails, you toss tails, and he tosses heads; or any sequence where both toss tails n times and then he tosses heads. So his probability of winning is at least $1/2 + 1/8 + 1/32 + \dots = 2/3$. So it seems you are better off moving away from the goal.

Rubric:

- 2 points for correct decision
- 4 points for justification

4. Consider that you live in the city of Townsville. The city is quite safe having the chance of a burglary at 0.1% and the chance of an earthquake at 0.2%. Your home is equipped with a burglar alarm that is reliable in detecting burglary, but it may be triggered by minor earthquakes. Specifically, the alarm goes off 94% of the time when there is a burglary, and this increases to 95% when an earthquake occurs simultaneously. Additionally, the alarm is triggered 29% of the time during an earthquake, even if there is no burglary. Without either an earthquake or burglary, there is a 0.1% chance that the alarm will go off randomly.

Your neighbors, Jamal and Munira, have agreed to notify you at work if they hear the alarm. Jamal calls 90% of the time when he hears the alarm, but occasionally (5%) mistakes the telephone ring for the alarm. Munira, due to nearby construction noise, might miss the alarm, but calls you 70% of the time when the alarm rings. She also calls 1% of the time even when the alarm does not go off.

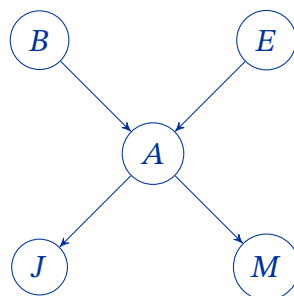
Assume that the probabilities summarize a potentially infinite set of circumstances that are not mentioned here.

- a) Formulate the scenario as a Bayesian Network considering causal relations and construct the associated Conditional Probability Tables.

20
(CO2)
(PO2)

Solution:

Let B = Burglary, E = Earthquake, A = Alarm, J = Jamal Calls, and M = Munira Calls. We have the following Bayesian Network:



The associated CPTs are as follows:

$P(B)$	
$+b$	0.001
$-b$	0.999

$P(E)$	
$+e$	0.002
$-e$	0.998

$P(A B, E)$			
$+a$	$+b$	$+e$	0.950
$+a$	$+b$	$-e$	0.940
$+a$	$-b$	$+e$	0.290
$+a$	$-b$	$-e$	0.001
$-a$	$+b$	$+e$	0.050
$-a$	$+b$	$-e$	0.060
$-a$	$-b$	$+e$	0.710
$-a$	$-b$	$-e$	0.999

$P(J A)$		
$+j$	$+a$	0.90
$+j$	$-a$	0.05
$-j$	$+a$	0.10
$-j$	$-a$	0.95

$P(M A)$		
$+m$	$+a$	0.70
$+m$	$-a$	0.01
$-m$	$+a$	0.30
$-m$	$-a$	0.99

Rubric:

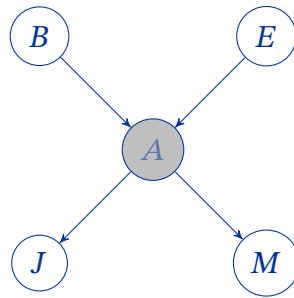
- 1 point for each correct node
- 1 point for each correct edge
- 1 point for no wrong edge/node
- 0.5 points for each probability entry

b) Identify and prove at least two conditional independence from your Bayesian Network.

Solution:

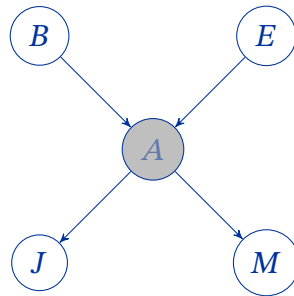
For $M \perp\!\!\!\perp B, E|A$, we have:

6
(CO3)
(PO3)



Here, the paths from B to M and E to M have one single triple each. Each of the triples are inactive. So the paths are inactive. Which makes M conditionally independent of B and E given A .

For $J \perp\!\!\!\perp B, E | A$, we have:



Here, the paths from B to J and E to J have one single triple each. Each of the triples are inactive. So the paths are inactive. Which makes J conditionally independent of B and E given A .

Rubric:

- 1 point for each condition independence
- 2 points for each proof.

- c) Find the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both Jamal and Munira called you.

2
(CO1)
(PO1)

Solution:

$$\begin{aligned}
 P(+j, +m, +a, -b, -e) &= P(+j | +a)P(+m | +a)P(+a | -b \wedge -e)P(-b)P(-e) \\
 &= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 \\
 &= 0.000628
 \end{aligned}$$

Rubric:

- 1 point for the correct equation
- 1 point for the result

5. a) Imagine a game between two players Pascal and Fermat where each turn consists of the roll of a fair 6-sided dice. On each turn, the dice is rolled and the face that lands as the top of the dice is considered as outcome of that turn. Pascal gets a point when the outcome is even and Fermat gets a point when the outcome is odd. The first player to get T points wins the game and is rewarded M USD. Suppose the game is interrupted with Pascal leading by $X - Y$, where $0 \leq Y < X < T$. Considering the game must end prematurely, determine the general formula on how a rational agent should fairly split the reward.

7
(CO1)
(PO1)

Solution:

Let p be the probability that Pascal wins each round. For a fair dice, $p = 0.5$. We can write:

$$m_{\text{Pascal}} = \begin{cases} 1 & \text{if } X = T \\ 0 & \text{if } Y = T \\ p \cdot m_{\text{Pascal}}(p, X + 1, Y, T) + (1 - p) \cdot m_{\text{Pascal}}(p, X, Y + 1, T) & \text{otherwise} \end{cases}$$

Pascal should be rewarded $M \times m_{\text{Pascal}}$ and Fermat should get $(M(1 - m_{\text{Pascal}}))$ USD.

Rubric:

- 1 point for the correct probability assumption.
- 1 point for each base case
- 2 points for the recursive case
- 1 point for each reward.

- b) Assume that you are building a rational diagnosis robot for a hospital that screens patients and assigns them doctors and/or tests based on observed symptoms. Based on the past history of the hospital, you have observed that 1 out of every 10000 patients admitted in the hospital have meningitis. You know that the disease meningitis causes a patient to have a stiff neck 80% of the time. Again, a person not having meningitis can have a stiff neck 1% of the time. A test for meningitis requires the collection of cerebrospinal fluid via a spinal tap, which is expensive.
- i. Suppose that a patient reports that they have a stiff neck. Should the robot send them for the test? Justify your choice.

9 + 7
(CO3)
(PO3)

Solution:

Here,

$$P(+m) = \frac{1}{10000} = 0.0001$$

$$P(+s | +m) = 0.8$$

$$P(+s | -m) = 0.01$$

We get:

$$\begin{aligned} P(+m | +s) &= \frac{P(+s | +m)P(+m)}{P(+s)} \\ &= \frac{P(+s | +m)P(+m)}{P(+s | +m)P(+m) + P(+s | -m)P(-m)} \\ &= 0.00794 \end{aligned}$$

We expect only 0.7% patients with a stiff neck to have meningitis. Even though a stiff neck is quite strongly indicated by meningitis, the probability of meningitis in patients with stiff necks remains small. This is because the prior probability of stiff necks is much higher than the prior probability for meningitis. So the robot should not send the patient for tests.

Rubric:

- 3 point for givens
- 2 point for equation and elaboration

- 1 point for result
- 3 points for justification

ii. Is there any reason to disagree with your choice in Question 5.b)i? Provide brief arguments in support of your decision.

Solution:

A patient should still consider getting checked. Because the utility of having meningitis and not getting checked (thus it being undetected) is lower than the utility of not having meningitis and getting checked (thus being relieved).

Rubric:

- 2 points for the decision.
- 5 points for the justification.